

Foundations Notebook

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June 8, 2026

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Chapter 1

Mathematical Foundations

1.1 Calculus

1.1.1 Vector Calculus

Coordinate Systems

A coordinate system is a method of describing a space that function exists in. The focus will be on that of three dimensional coordinate systems; many examples exist:

1. Cartesian(x,y,z): A point is specified by its coordinates along each basis vector relative to the origin (see Section ??).
2. Cylindrical(ρ, ϕ, z): A point is defined by its radial distance from the origin(ρ), angle relative to the x direction (ϕ), and its z coordinate.

$$x = \rho \cos \phi, \quad (1.1)$$

$$y = \rho \sin \phi, \quad (1.2)$$

$$z = z. \quad (1.3)$$

$$\rho = \sqrt{x^2 + y^2}, \quad (1.4)$$

$$\phi = \text{atan2}(y, x), \quad (1.5)$$

$$z = z. \quad (1.6)$$

3. Spherical(ρ, θ, ϕ): A point is defined by its radial distance from the origin (ρ) and its angle relative to $x(\phi)$ and $z(\theta)$.

$$x = \rho \sin \theta \cos \phi, \quad (1.7)$$

$$y = \rho \sin \theta \sin \phi, \quad (1.8)$$

$$z = \rho \cos \theta \quad (1.9)$$

$$\rho = \sqrt{x^2 + y^2 + z^2}, \quad (1.10)$$

$$\phi = \text{atan2}(y, x), \quad (1.11)$$

$$\theta = \arccos\left(\frac{z}{r}\right). \quad (1.12)$$

Vectors

A vector is a quantity that possesses both magnitude and direction and is commonly denoted as follows:

$$\mathbf{v} = \langle a, b \rangle. \quad (1.13)$$

Here, the vector $\mathbf{v}$ represents a vector originating at the origin and terminating at the point (a, b) .

Vector operations include addition, subtraction, multiplication with a scalar, dot product, and cross product.

1. Vector Addition: Adding vectors involves summing the corresponding components of each vector. That is,

$$\langle x_1, y_1, z_1 \rangle + \langle x_2, y_2, z_2 \rangle = \langle x_1 + x_2, y_1 + y_2, z_1 + z_2 \rangle \quad (1.14)$$

2. Vector Subtraction: Similar to that of vector addition with the exception that one vector has its direction reversed.
3. Scalar Multiplication: Multiplying a vector by a scalar scales its magnitude; it changes the vector's length but does not change its direction, except that a negative scalar reverses the direction.
4. Dot Product:

Equation of a Line

The equation of a line in a 2 dimensional space is as follows

$$y = a * x + b, \quad (1.15)$$

where a is the slope or the ratio of change relating a change in x to a change in y and b is a starting value of y when x is at the origin.

In order to calculate a line in 3 dimensions three components are needed: a known point on the line($\vec{a}_0 = \langle x_0, y_0, z_0 \rangle$), a arbitrary point on the line($\vec{a} = \langle x, y, z \rangle$) and a vector parallel to the line($\vec{v}$). With these components and the vector operation noted in Section 1.1.1 the equation is as follows:

$$\vec{b} = \vec{a}_0 - \vec{a}, \quad (1.16)$$

$$\vec{a} = \vec{a}_0 + \vec{b}, \quad (1.17)$$

$$\vec{a} = \vec{a}_0 + t * \vec{v}. \quad (1.18)$$

Such that, t is a arbitrary scalar multiplied with the parallel vector $\vec{v}$. This can be further expanded to

$$\vec{a} = \vec{a}_0 + t * \vec{v}, \quad (1.19)$$

$$\vec{a} = \langle x_0, y_0, z_0 \rangle + t * \langle x_v, y_v, z_v \rangle, \quad (1.20)$$

$$\vec{a} = \langle x_0 + t * x_v, y_0 + t * y_v, z_0 + t * z_v \rangle \quad (1.21)$$

Note that this equation will generate a vector that points to all points on the line as determined by the value of t and can be thought of as a laser pointer following the line.

Equation of a Plane

Flux

1.2 Linear Algebra

1.2.1 Vector Spaces

1.2.2 Why This Matters

Linear algebra underpins signal spaces, optimization, and estimation theory.

Chapter 2

Electrical Engineering Foundations

2.1 Electro Magnetism

2.1.1 coulombs Law

2.1.2 Maxwells Equations

$$\nabla \cdot \mathbf{D} = \rho, \quad (2.1)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (2.2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad (2.3)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \quad (2.4)$$

where $\mathbf{E}$ is the electric field intensity in the units of V m^{-1} , $\mathbf{H}$ is the magnetic field intensity in the units of A m^{-1} , $\mathbf{D}$ is the electric flux (see section 1.1.1) densities in the units of C m^{-2} , and $\mathbf{B}$ is the magnetic flux density in the units of Wb m^{-2} or a Tesla(T), $\mathbf{D}$ is the electric displacement and (B) is magnetic induction.

2.1.3 Electric Field

This type of electric field is based static charge, and can be thought of like gravity such that the force being exerted is all around if in a 3 d manner that diminishes with the greater range from the source.

$$F = \frac{1}{4\pi\epsilon} \left(\frac{Q_1 Q_2}{r^2} \right) \quad (2.5)$$

2.2 Radar

Radar: Radio Detection And Ranging. Is simply the transmission and receiving of Radio waves. From these radio waves it is possible to determine many things about the incident target.

Simple Model

A transmitter is to be thought as a isotropic radiator with the propagation pattern described by equation 2.6.

$$P_d = \frac{P_t}{4\pi r^2} \quad (2.6)$$

Where P_d is the power density at some radius, P_t is the power transmitted and r is the radius.

$$R = \frac{c\delta t}{2} \quad (2.7)$$

2.2.1 SAR

2.3 Signals and Systems

2.3.1 Adaptive Filtering

What is an Adaptive filter

An adaptive filter uses a Wiener Filter structure but iterates it and adjusts it such that the error is reduced and it begins to approach the ideal filter.

$$h(n+1) = h(n) + \Delta h(n) \quad (2.8)$$

$$\Delta h(n) = -\gamma \left(\frac{d\sigma_{e_{\min}}^2}{dh} \right) \quad (2.9)$$

Applications

Using the ideas of Wiener Filters and Error Surfaces they can be applied in the ideas of signal recovery in a noisy channel and noise cancellation.

Noise cancellation Given a the noisy signal and correlated noise we are able to filter the correlated noise such that it matches the noise of the signal and thus allows us to cancel it out leaving only the signal in its place.

2.3.2 Sampling

2.3.3 LTI Systems

An LTI system satisfies:

- Linearity
- Time invariance

2.3.4 Idea

- Optical PCB's

Chapter 3

Music Theory

3.1 Structure

3.1.1 Scale

A scale is division of a octave into a collection of notes. The two primary classes of scales are major and minor.

1. Major: W,W,H,W,W,W,H
2. Minor: W,H,W,W,H,W,W

3.1.2 Key

Key can be thought of as the root note or tonic of a scale. To shift between keys a commonality is needed for a smooth transition whether it be note or chord. For instance C major to G major; see Figure 3.1 as reference between key transitions.

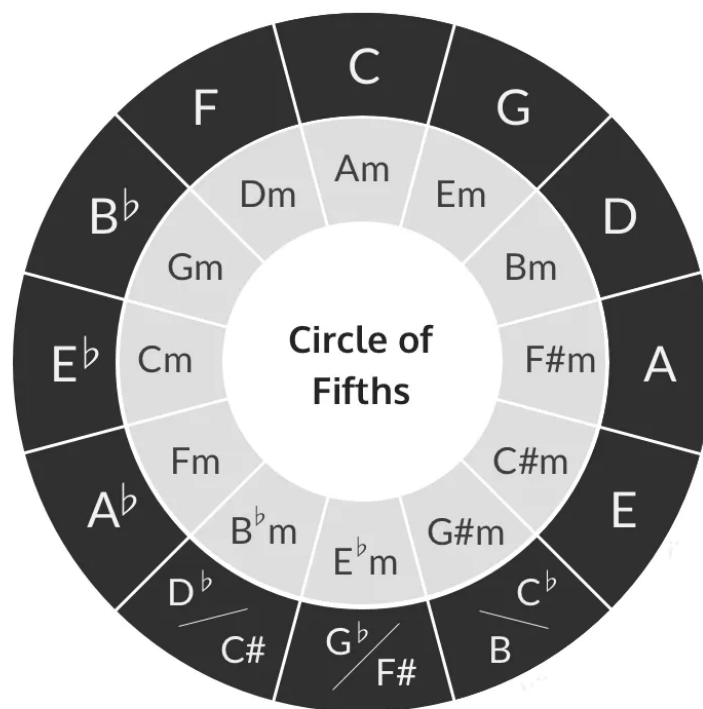


Figure 3.1: Circle of fifths